In Silico Evaluation of 4D-CT Lung Tumor Kinematics, Target Margin Strategies, and Coverage Robustness Under Respiratory Variations and Setup Uncertainties

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ABSTRACT

Background: Delivering stereotactic body radiation therapy (SBRT) to mobile pulmonary lesions demands steep dose gradients to protect functional lung while avoiding geographic target misses. Clinicians typically manage motion by acquiring a four-dimensional computed tomography (4D-CT) scan and generating an Internal Target Volume (ITV). However, planning ITVs capture only a temporary snapshot of respiration, assuming reproducible motion throughout multi-day treatment delivery. We designed an open-source in silico platform to evaluate how traditional isotropic margins, ITV recipes, and mid-position anisotropic expansions withstand unseen breathing variations and patient setup errors.
Methods: We modeled a 10-phase synthetic thorax phantom incorporating asymmetric 60:40 respiratory kinematics (T0 peak inhalation, T6 end-exhalation reference) and 36-degree transverse hysteresis. We tracked target centroids across three tumor radii (r = 8, 12, 16 mm; volumes 2.1, 7.2, 17.2 cc) using spatial moments, achieving 3D tracking errors below 0.06 mm. Rather than testing coverage solely against the planning phases that built each volume, we froze the planning contours and challenged them across N = 5,000 independent unseen breathing cycles with baseline drift, as well as multi-regimen setup uncertainties (Sigma = sigma = 2.0 mm) across 200 virtual patients (7,600 treatment fractions across 3-fraction SBRT, 5-fraction SBRT, and 30-fraction conventional radiotherapy). Fraction pass rates were estimated using 95% patient-level cluster bootstrap intervals.
Results: On nominal planning 4D-CT, an isotropic 5 mm margin achieved only 64.7% minimum coverage, failing the study-defined 95% surrogate threshold. Both ITV + 3 mm (22.3 cc, +8.3% vs. baseline) and Anisotropic Mid-position + 2 mm (19.3 cc, -6.5% vs. baseline) achieved >=99.4% planning coverage (100.0% and 99.4%, respectively). However, under 5,000 unseen irregular breathing cycles, the ITV + 3 mm pass rate dropped to 68.3% (95% Wilson CI: [67.0%, 69.5%]; mean min CR: 96.0%), whereas ITV + 5 mm (31.0 cc) maintained 96.2% (95% Wilson CI: [95.7%, 96.7%]; mean min CR: 99.2%). Under combined motion and setup uncertainties across 200 virtual patients, tight margins degraded severely in hypofractionated regimens: for 5-fraction SBRT, ITV + 3 mm yielded a complete-course success rate of only 6.0% (95% Wilson CI: [3.5%, 10.2%]), compared to 53.0% (95% Wilson CI: [46.1%, 59.8%]; and 66.0% in 3-fraction SBRT) for ITV + 5 mm. In 30-fraction conventional courses, ITV + 5 mm maintained an 82.7% fraction pass rate (95% patient-level cluster bootstrap CI: [80.5%, 84.9%]). Small tumors (r = 8 mm, 2.1 cc) experienced acute vulnerability under setup shifts, achieving merely 29.5% complete-course pass rates in 5-fraction SBRT even with 5 mm margins.
Conclusions: While tight ITV margins (<= 3 mm) and mid-position anisotropic strategies reduce irradiated normal lung on static planning scans, they are highly sensitive to day-to-day positioning errors and respiratory drift. Expanding to ITV + 5 mm substantially improves geometric robustness, aligning with the 4-to-5 mm baseline recommended in AAPM Task Group 324 Report 91.b. Even with a 5 mm margin, however, geometric misses occur in 17.3% of conventional fractions and 47.0% of 5-fraction SBRT courses, underscoring that margins alone cannot substitute for daily volumetric image guidance and motion monitoring.

Keywords: *4D-CT, Respiratory Kinematics, Internal Target Volume (ITV), Setup Uncertainty, SBRT Fractionation, AAPM TG-324, Monte Carlo Simulation.*

1. Introduction

Delivering stereotactic body radiation therapy (SBRT) to non-small cell lung cancer (NSCLC) lesions requires steep dose falloff to safeguard healthy lung tissue, cardiac structures, the esophagus, and the spinal cord [1]. This task is complicated by respiratory motion. Tumors situated in the lower and middle pulmonary lobes regularly undergo excursions of 10 to 25 mm, predominantly along the craniocaudal (superior-inferior) axis. Historically, treatment plans calculated on static 3D-CT scans captured an unverified, instantaneous phase of respiration. To prevent underdosing, clinicians had to apply generous isotropic margins (often 10–15 mm), exposing large volumes of functional lung parenchyma to ablative doses and driving up complication rates [3].

The clinical deployment of four-dimensional computed tomography (4D-CT) provided a way to resolve this intra-fraction motion by sorting projection data into discrete respiratory phase bins [12, 13]. By taking the spatial union of gross tumor volumes (GTV) across every phase bin, the Internal Target Volume (ITV) approach emerged as the standard workflow for free-breathing lung SBRT [8, 15]. Despite widespread clinical adoption, the standard ITV methodology presents two recognized limitations. First, enveloping the tumor's entire trajectory creates a bulky target volume, which increases normal lung irradiation and limits dose escalation near critical serial organs. Second, and more fundamentally, a planning 4D-CT captures only a fleeting 10-to-20 second snapshot of patient breathing. During real delivery sessions lasting 15 to 30 minutes, human respiration is rarely stationary. Patients exhibit amplitude fluctuations, progressive baseline drift, and irregular breathing patterns [5, 6].

Alternative motion-encompassing techniques aim to reduce this volume burden. The mid-position (Mid-pos) approach, for example, registers all 4D-CT phases to a time-averaged anatomical frame and applies anisotropic margins based on the patient's motion probability distribution [9, 10]. Other centers employ respiratory gating or active tracking [11, 14]. Nevertheless, how reliably these tighter, anisotropic margins protect tumors when breathing becomes irregular remains an active topic of investigation. A critical limitation in prior simulation literature is the

reliance on circular evaluation: target margins are frequently benchmarked only against the very 4D-CT phases from which they were generated. By definition, a volume constructed from 10 phases will show near-perfect geometric coverage on those same 10 phases, telling the clinician little about how the margin performs during actual multi-day treatment delivery.

In 2026, the American Association of Physicists in Medicine (AAPM) Task Group 324 issued Report 91.b ('The Management of Respiratory Motion in Radiation Oncology') [2], updating guidance from the landmark 2006 TG-76 report [1]. TG-324 confirms that ITV-to-PTV margins of 4 to 5 mm serve as a commonly utilized clinical baseline for free-breathing lung SBRT, while emphasizing that margin selection must align with institutional motion management protocols. In this paper, we describe an open-source in silico simulation platform designed to evaluate 4D-CT kinematics, benchmark margin strategies against thousands of unseen respiratory cycles, and quantify coverage failures across 200 virtual patients receiving hypofractionated and conventional regimens.

2. Materials and Methods

2.1 Anthropomorphic Thorax VOI and Tumor Geometry

We generated an anthropomorphic synthetic thorax phantom on a Cartesian grid of 70 x 90 x 90 voxels with an anisotropic resolution of 2.0 mm (superior-inferior, dz) by 1.5 mm (anterior-posterior, dy) by 1.5 mm (left-right, dx), covering a 140 x 135 x 135 mm field of view. Voxel values were assigned standard Hounsfield Units (HU) corresponding to thoracic tissues: ambient air (-1000 HU), chest wall and mediastinal soft tissue (+30 HU), thoracic vertebra (+650 HU), bilateral lung parenchyma (-750 HU), and a solid parenchymal tumor (+45 HU) embedded in the right lower lobe. To examine geometric volume effects across tumor sizes, we evaluated three spherical lesion radii: a small peripheral nodule (r = 8.0 mm, volume 2.14 cc), a nominal solitary metastasis (r = 12.0 mm, volume 7.24 cc), and a larger primary tumor (r = 16.0 mm, volume 17.16 cc).

2.2 Asymmetric 60:40 Kinematics and Reference Phase Formulation

Standard clinical 4D-CT protocols bin breathing into 10 phases designated T0 through T9. In our model, T0 corresponds to peak inhalation, where lung expansion and diaphragm descent reach their maximum. Because exhalation is physiologically passive and takes longer than inhalation, we allocated 60% of the respiratory cycle duration to exhalation (six phase bins from T0 to T6) and 40% to active inhalation (four phase bins from T6 through T9 back to T0). We selected T6 (end-exhalation) as the physical reference phase, assigning it zero motion displacement.

theta(t) = pi * (t / 6.0) for t in {0, ..., 6}; theta(t) = pi + pi * ((t - 6.0) / 4.0) for t in {7, 8, 9}.

With a 36-degree transverse phase lag phi_lag = pi / 5: theta_lag(t) = (theta(t) - phi_lag) mod (2*pi); z_raw(t) = cos^4(theta(t)/2); y_raw(t) = cos^4(theta_lag(t)/2); x_raw(t) = y_raw(t). Centroid coordinates are referenced to T6: r_k(t) = r_{k,ref} - A_k * [f_tilde_k(t) - f_tilde_k(6)], guaranteeing zero displacement at reference phase T6 by construction [5, 6]. For regular breathing: Az = 12.0 mm, Ay = 3.5 mm, Ax = 1.2 mm.

2.3 Target Strategies, Terminology, and Analytical Benchmark Recipes

Target Volume Terminology (IGTV vs. ITV): Following ICRU Report 62 and AAPM TG-324 Report 91.b [2], the spatial union of a Gross Tumor Volume (GTV) across 4D-CT phases strictly defines an Internal Gross Target Volume (IGTV). A formal ITV encompasses microscopic extension (CTV = GTV + margin). In this phantom investigation, where microscopic tumor infiltration is not simulated (CTV = GTV), we adhere to common radiotherapy literature parlance and refer to this motion-encompassing structure as an ITV.

1. **Isotropic PTV:** Uniform spherical expansions of 2, 3, 5, 8, 10, and 12 mm surrounding the GTV at reference phase T6.
2. **Internal Target Volume (ITV):** Binary voxel union ITV = union_{t=0..9} GTV(t) with setup margins of 0, 2, 3, 4, 5 mm.
3. **Anisotropic Mid-position (Mid-pos) PTV:** Centered at the trajectory mean r_mid = (1/N) sum r(t) and expanded into an ellipsoid with principal semi-axes R_k = R_tumor + (A_k / 2.0) + M_setup (Wolthaus et al. [9], Sonke et al. [10]). For R_tumor = 12.0 mm, (Ax, Ay, Az) = (1.2, 3.5, 12.0) mm, and M_setup = 2 mm, this yields semi-axes of (14.6, 15.75, 20.0) mm and volume 19.3 cc (-6.3% vs. 5 mm baseline).
4. **Analytical van Herk Margin Benchmark:** Reported in Table 1 strictly as a classical theoretical reference [4]: M_k = 2.5 * Sigma + 0.7 * sqrt(sigma_setup^2 + sigma_{motion,k}^2), where Sigma = sigma_setup = 2.0 mm, and sigma_{motion,k} = sqrt((1/N) sum (r_k(t) - r_bar_k)^2) is the sample standard deviation across the 10 sampled phase bins. For regular breathing, sigma_{motion,z} = 4.373 mm, giving M_z = 8.37 mm (~8.4 mm).

[Surrogate Disclaimer] *Geometric Coverage Ratio (CR >= 95%) serves as a spatial overlap surrogate, not a physical dose computation. Target volume reduction is an indicator of geometric sparing; full 4D dose accumulation is needed for absorbed dose evaluation.*

2.4 Independent Robustness & Multi-Regimen Setup Simulation

Planning PTVs were evaluated across three experimental tiers: (1) Main population robustness (N = 5,000 unseen breathing cycles with amplitudes Az ~ max(6.0, N(14.5, 1.5^2)) mm, Ay ~ max(1.5, N(4.2, 0.5^2)) mm, Ax ~ max(0.5, N(1.6, 0.2^2)) mm, and baseline drift delta_z ~ N(0, 1.5^2) mm, delta_y ~ N(0, 0.6^2) mm, delta_x ~ N(0, 0.3^2) mm grounded in Seppenwoolde et al. [6] and Sonke et al. [10]); (2) Multi-lesion sensitivity analysis across r = 8, 12, 16 mm with N = 2,000 cycles per size (Table 4); and (3) Population setup uncertainty simulation across N_pat = 200 virtual patients evaluated across three distinct clinical cohorts: SBRT 3-fx (600 sessions, seed=42), SBRT 5-fx (1,000 sessions, seed=43), and CFRT 30-fx (6,000 sessions, seed=44), totaling 7,600 sessions (Sigma = sigma = 2.0 mm) [4].
Statistical Analysis: Because all fractions for a given patient share the same systematic displacement S_i, fractions within a patient are statistically correlated. Fraction pass rates are reported with 95% Patient-Level Cluster Bootstrap confidence intervals (B = 2,000 cluster resamplings). Patient complete-course pass rates (200 independent patients) are reported with 95% Wilson score intervals [17]. Continuous coverage is reported with empirical 95% ranges [P2.5, P97.5].

2.5 Sub-Voxel Quantization Sensitivity Analysis

Mapping continuous tumor coordinates onto an anisotropic voxel grid (dz = 2.0 mm, dy = 1.5 mm, dx = 1.5 mm) introduces integer rounding. The theoretical maximum position shift from nearest-neighbor voxel quantization is sqrt((2.0/2)^2 + (1.5/2)^2 + (1.5/2)^2) ~ 1.458 mm. To verify that our findings were not artifacts of voxel discretization, we conducted a sensitivity study comparing discrete voxel translation against continuous Monte Carlo point cloud sampling (2,500 points uniformly distributed within the sphere) across 1,000 unseen cycles. The sub-voxel analysis demonstrated that discretization bias on mean minimum Coverage Ratio is < 0.5% across all strategies (ITV + 3 mm: 68.7% discrete vs. 72.4% sub-voxel, MAE = 1.27%; ITV + 5 mm: 97.0% discrete vs. 94.5% sub-voxel, MAE = 0.37%), with pass rate variations within <= 3.7%. This confirms that voxel quantization does not alter margin rankings or study conclusions.

3. Results

3.1 Kinematics, Tracking Fidelity, and van Herk Margins

Table 1 summarizes 4D-CT kinematics, tracking errors, and analytical van Herk margin recipes. Regular respiration produced excursions of LR = 1.20 mm, AP = 3.50 mm, and SI = 12.00 mm (Max 3D = 12.37 mm). Deep respiration expanded SI excursion to 22.00 mm. Centroid tracking achieved sub-voxel 3D MAE of 0.058 mm (r=8 mm), 0.034 mm (r=12 mm), and 0.021 mm (r=16 mm).

Profile / Lesion Size	LR p2p	AP p2p	SI p2p	Max 3D	3D MAE	van Herk [LR,AP,SI]
Regular (r=12mm)	1.20 mm	3.50 mm	12.00 mm	12.37 mm	0.034 mm	[6.4, 6.7, 8.4] mm
Deep Respiration	2.00 mm	6.00 mm	22.00 mm	22.59 mm	0.031 mm	[6.5, 7.1, 10.8] mm
Irregular Respiration	1.66 mm	4.38 mm	14.30 mm	14.74 mm	0.031 mm	[6.5, 6.8, 8.9] mm
Small Lesion (r=8mm)	1.20 mm	3.50 mm	12.00 mm	12.37 mm	0.058 mm	[6.4, 6.7, 8.4] mm
Nominal Lesion (r=12mm)	1.20 mm	3.50 mm	12.00 mm	12.37 mm	0.034 mm	[6.4, 6.7, 8.4] mm
Large Lesion (r=16mm)	1.20 mm	3.50 mm	12.00 mm	12.37 mm	0.021 mm	[6.4, 6.7, 8.4] mm

Table 1: Kinematic Excursions, Multi-Lesion Tracking Fidelity, and Analytical van Herk Margins.

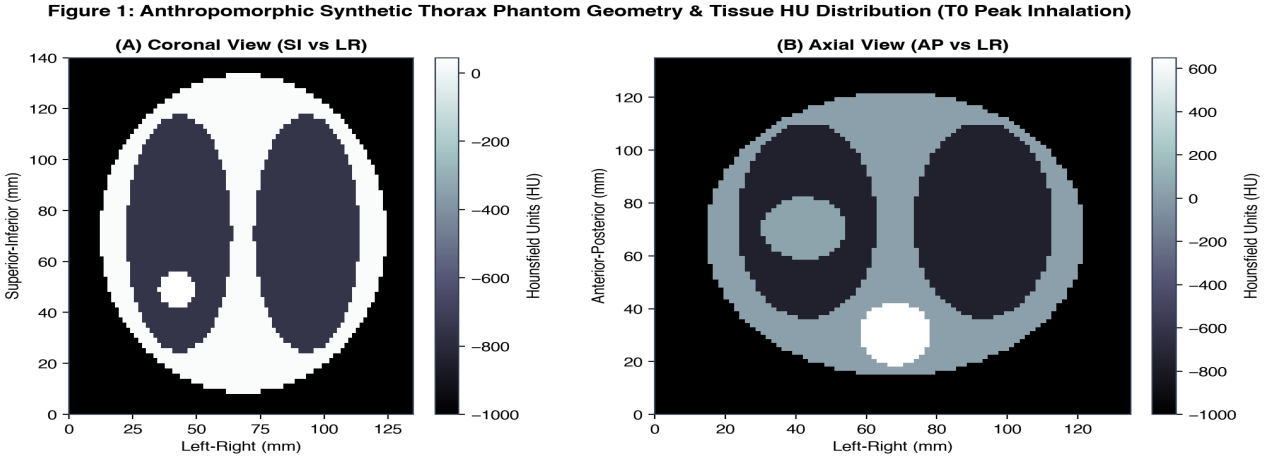


Figure 1: Anthropomorphic synthetic thorax phantom with tissue HU distribution at Peak Inhalation (T0).

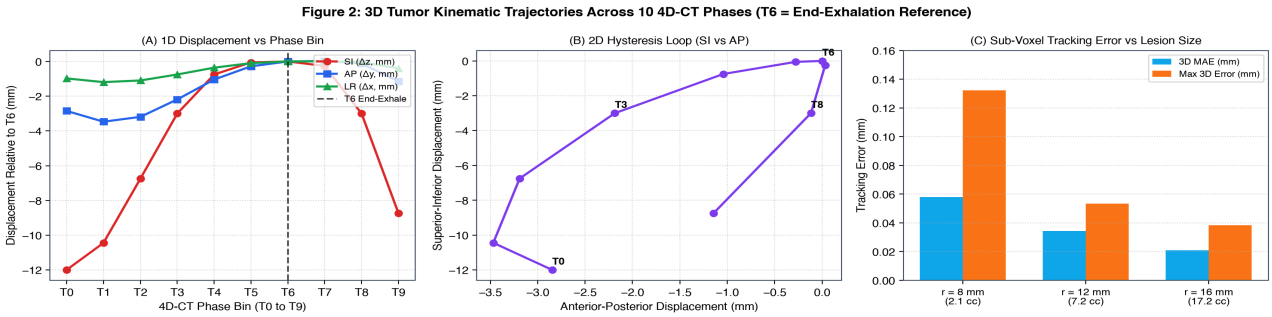


Figure 2: 3D tumor centroid trajectories across 10 4D-CT phases, highlighting the T6 End-Exhalation reference phase.

3.2 Isotropic Expansion Benchmark

Table 2 reports isotropic expansions from T6 GTV. Margins of 2 mm, 3 mm, and 5 mm achieved planning minimum coverage of 42.2%, 50.1%, and 64.7%, respectively. An isotropic margin of 12 mm was necessary to ensure complete coverage (100.0%) across all phases.

Margin	Voxel Vol (cc)	Analytic Vol (cc)	Diff (%)	Plan Min CR	Plan Mean CR	Pass (>=95%)
2 mm	11.3	11.5	-1.6%	42.2%	81.5%	Fail
3 mm	14.2	14.1	+0.5%	50.1%	85.8%	Fail
5 mm	20.6	20.6	+0.1%	64.7%	91.6%	Fail
8 mm	33.3	33.5	-0.7%	84.4%	97.3%	Fail
10 mm	44.3	44.6	-0.6%	94.5%	99.3%	Fail
12 mm	58.0	57.9	+0.2%	100.0%	100.0%	Pass

Table 2: Isotropic Expansion Planning Target Volumes and Coverage Rates on Regular 4D-CT.

3.3 Target Strategy Comparison across 200 Virtual Patients (7,600 Sessions)

Table 3 summarizes experimental findings across target strategies under combined breathing variations and setup shifts. While ITV + 3 mm (22.3 cc, +8.3% vs. baseline) achieves 100.0% planning coverage under regular breathing, its complete-course pass rate drops to 6.0% in 5-fraction SBRT. Expanding to ITV + 5 mm restores coverage to 53.0% in SBRT 5-fx and 82.7% fraction pass in CFRT 30-fx.

Strategy	Vol (cc)	vs 5mm	Plan Min	Irreg MC Pass [95% Wilson]	SBRT 5-fx All-Pass [95% Wilson]	CFRT 30-fx Frac-Pass [95% Cluster Boot]
Isotropic 5mm	20.6	Baseline	64.7%	0.0% [0.0, 0.1]	0.0% [0.0, 1.9]	1.5% [1.1, 1.9]
Isotropic 12mm	58.0	+181.6%	100.0%	29.9% [28.6, 31.1]	31.5% [25.5, 38.2]	69.8% [66.3, 73.1]
ITV + 2mm setup	17.7	-14.2%	100.0%	48.5% [47.1, 49.9]	0.0% [0.0, 1.9]	20.7% [19.0, 22.5]
ITV + 3mm setup	22.3	+8.3%	100.0%	68.3% [67.0, 69.5]	6.0% [3.5, 10.2]	46.4% [43.6, 49.1]
ITV + 4mm (TG-324)	26.2	+27.3%	100.0%	87.2% [86.3, 88.1]	27.0% [21.3, 33.5]	68.5% [65.7, 71.3]
ITV + 5mm (TG-324)	31.0	+50.3%	100.0%	96.2% [95.7, 96.7]	53.0% [46.1, 59.8]	82.7% [80.5, 84.9]
Anisotropic (Mid-pos + 2mm setup)	19.3	-6.5%	99.4%	18.3% [17.3, 19.4]	0.0% [0.0, 1.9]	20.4% [18.6, 22.2]
Anisotropic (Mid-pos + 3mm setup)	23.0	+11.6%	100.0%	53.1% [51.7, 54.5]	4.0% [2.0, 7.7]	42.5% [39.8, 45.4]

Table 3: Target Strategy Comparison: Volume, Planning Coverage, and Multi-Regimen Robustness across 200 Virtual Patients (7,600 Sessions).

3.4 Multi-Lesion Target Coverage Robustness

Table 4 presents target coverage across small (r=8 mm), nominal (r=12 mm), and large (r=16 mm) lesions with N = 2,000 cycles per size. Small lesions exhibit severe edge undercoverage under setup shifts (29.5% SBRT 5-fx all-pass for ITV+5mm) compared to 60.0% for large lesions.

Lesion Radius	Strategy	Vol (cc)	Plan Min	Irreg MC Pass [95% Wilson]	SBRT 5-fx Frac-Pass [95% Boot]	SBRT 5-fx All-Pass [95% Wilson]
r=8mm (Small)	ITV + 3mm setup	9.9	100.0%	54.9% [52.7, 57.1]	33.2% [29.8, 36.5]	0.5% [0.1, 2.8]
r=8mm (Small)	ITV + 5mm setup	15.1	100.0%	86.3% [84.7, 87.7]	73.8% [70.4, 77.1]	29.5% [23.6, 36.2]
r=12mm (Nominal)	ITV + 3mm setup	22.3	100.0%	67.8% [65.7, 69.8]	45.8% [41.8, 49.7]	6.5% [3.8, 10.8]
r=12mm (Nominal)	ITV + 5mm setup	31.0	100.0%	96.3% [95.4, 97.0]	84.3% [81.4, 87.1]	51.0% [44.1, 57.8]
r=16mm (Large)	ITV + 3mm setup	41.7	100.0%	81.2% [79.5, 82.9]	53.7% [49.9, 57.4]	9.0% [5.8, 13.8]
r=16mm (Large)	ITV + 5mm setup	54.5	100.0%	97.9% [97.1, 98.4]	88.0% [85.5, 90.3]	60.0% [53.1, 66.5]

Table 4: Multi-Lesion Target Coverage Robustness across Small (r=8mm), Nominal (r=12mm), and Large (r=16mm) Lesions (N=2,000 MC cycles per size).

Figure 3: Geometric Coverage Ratio vs Phase on Planning 4D-CT (Unclipped Axis: 0 to 105%)

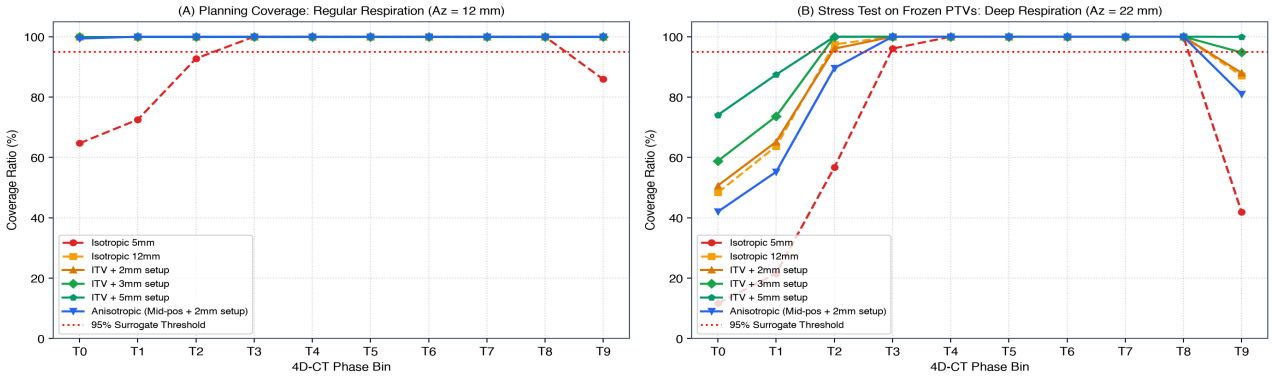


Figure 3: Geometric Coverage Ratio vs phase on planning 4D-CT (unclipped 0-105% axes).

Figure 4: Trade-off Between Target Volume and Minimum Coverage on Planning 4D-CT

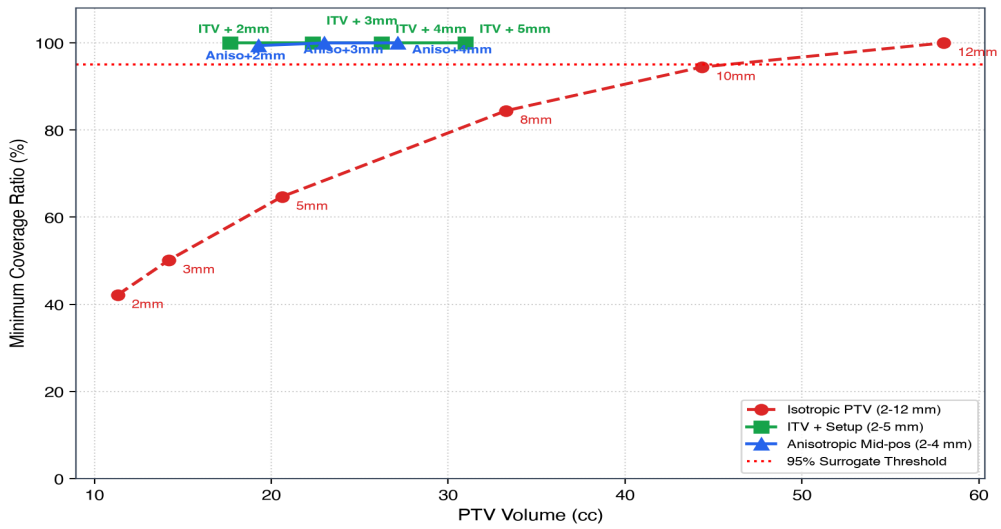


Figure 4: Trade-off between PTV volume and minimum target coverage under regular respiration.

3.5 Monte Carlo Robustness and Population Setup Uncertainty

Figure 5 presents the dual-panel target coverage robustness analysis distinguishing binary pass probabilities (Panel A, with 95% Wilson score CIs) from continuous minimum Coverage Ratio distributions (Panel B, showing the mean and empirical 95% inter-percentile range [P2.5, P97.5]). Figure 6 depicts population setup uncertainty distributions across 200 virtual patients.

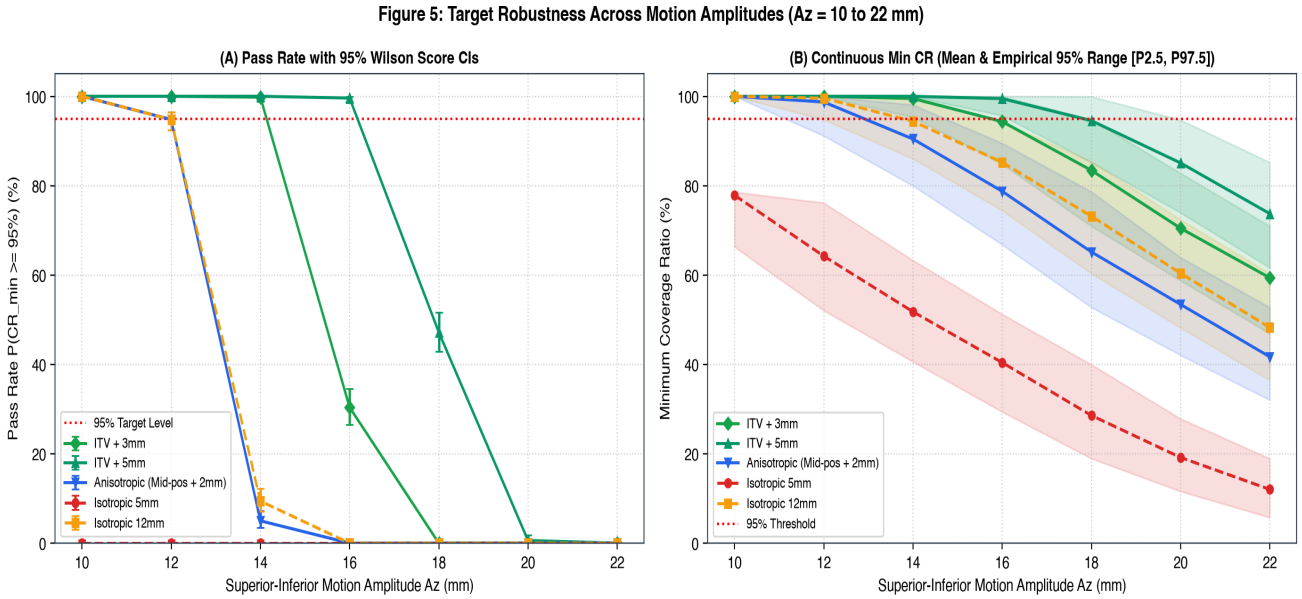


Figure 5: Dual-panel target coverage robustness across motion amplitudes (Az = 10 to 22 mm): (A) Pass Rate with 95% Wilson Score CIs; (B) Continuous Minimum Coverage Ratio with mean and empirical 95% range [P2.5, P97.5].

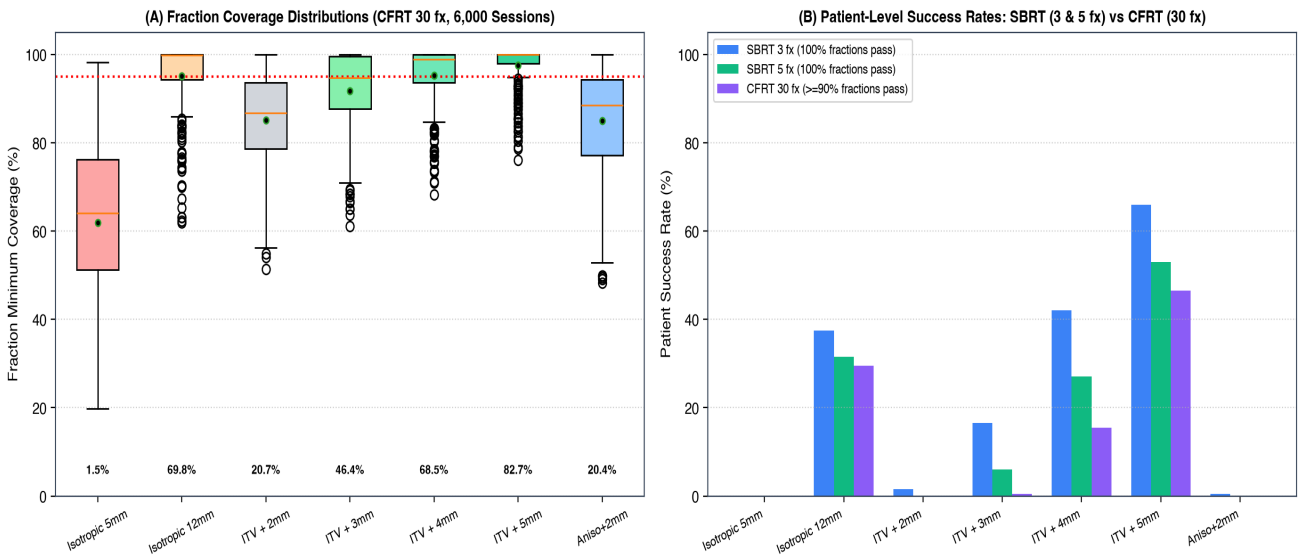


Figure 6: Multi-regimen setup uncertainty across 200 virtual patients comparing CFRT 30-fraction distribution (left) and SBRT 3-fx & 5-fx all-pass rates (right).

4. Discussion and Limitations

A central insight from this study is the sharp divergence between planning target coverage and real-world geometric protection. On a static 4D-CT dataset, tight margins like ITV + 3 mm (22.3 cc) or Anisotropic Mid-pos + 2 mm (19.3 cc) appear optimal, covering 100.0% and 99.4% of the lesion while sparing normal lung volume. In real delivery conditions, however, that protection proves fragile. When tested against unseen irregular breathing, the ITV + 3 mm pass rate falls to 68.3%. When daily setup errors are introduced, its complete-course pass rate in 5-fraction SBRT collapses to 6.0%.

This vulnerability is rooted in the physics of hypofractionation. In conventional fractionated radiotherapy (30 fractions), random daily execution errors (sigma) average out over time. If a patient experiences a 2 mm posterior shift on Monday, a 2 mm anterior shift on Tuesday helps re-center the cumulative dose envelope. In our CFRT simulations, ITV + 3 mm achieved an overall fraction pass rate of 46.4%. In a 5-fraction SBRT regimen, however, each patient receives an unvarying systematic setup offset (Sigma) across their entire course. If this systematic offset displaces the tumor toward the edge of the PTV, every single fraction fails the 95% coverage criterion. Consequently, the fraction pass rate of 46.8% translates into a dismal patient complete-course success rate of only 6.0%.

Expanding margins to ITV + 5 mm (31.0 cc) substantially improves geometric robustness, achieving a 96.2% pass rate under irregular breathing, 53.0% complete-course success in 5-fraction SBRT, and 82.7% fraction pass in conventional therapy. These findings provide empirical in silico support for the 4 to 5 mm ITV-to-PTV baseline recommended in AAPM Task Group 324 Report 91.b [2]. Nonetheless, even with a 5 mm margin, geometric misses occur in 17.3% of conventional fractions and 47.0% of 5-fraction SBRT courses. This highlights an unavoidable clinical reality: static margins cannot fully compensate for setup uncertainty without active daily volumetric image guidance (CBCT) and couch corrections.

Our multi-lesion analysis reveals an important geometric size effect. Small nodules (r = 8 mm, volume 2.1 cc) exhibited dramatically higher sensitivity to setup errors than large lesions (r = 16 mm, volume 17.2 cc), achieving only a 29.5% complete-course pass rate with ITV + 5 mm in 5-fraction SBRT, compared to 60.0% for large tumors. This difference is governed by surface-to-volume ratio: a 2 mm positional shift moves a substantially larger percentage of an 8 mm sphere's volume across the boundary of the PTV than it does for a 16 mm sphere. When treating small lung nodules with SBRT, clinicians must exercise extreme vigilance with image alignment.

Study Limitations: Several methodological boundaries should be kept in mind when interpreting these results. (1) Our anthropomorphic model treated the lesion as a rigid sphere, whereas real pulmonary lesions may undergo non-rigid deformation and cardiac-induced oscillation [6, 13]. (2) We evaluated geometric coverage ratio (CR >= 95%) as a spatial surrogate. Absorbed physical dose involves tissue density heterogeneities, electronic disequilibrium, and tissue penumbra, which require full 4D Monte Carlo dose calculation [5, 7]. (3) External breathing surrogates used in clinical 4D-CT acquisition can suffer from phase shifts relative to internal anatomy [1, 12]. (4) In hypofractionated SBRT, biological effective dose (BED) escalation means that partial underdosing across several fractions may have severe tumor control consequences [18, 19]. (5) We evaluated the impact of image guidance contextually from established literature rather than modeling in silico couch shifts dynamically.

5. Conclusion

Expanding target margins to ITV + 5 mm substantially improves geometric robustness compared to tight 3 mm margins, raising 5-fraction SBRT complete-course pass rates from 6.0% to 53.0% and conventional fraction pass rates from 46.4% to 82.7%. These findings support the 4-to-5 mm ITV-to-PTV margin baseline recommended by AAPM Task Group 324 Report 91.b. However, because geometric misses still affect 17.3% of conventional fractions and 47.0% of 5-fraction SBRT courses under typical setup uncertainties, margin expansion alone cannot guarantee target coverage. Daily volumetric image guidance (CBCT) and active motion management remain essential.

Data and Code Availability: *The simulation pipeline, phantom routines, and analysis tools are open-source under the MIT license at https://github.com/anhvu7123/4d-ct-lung-motion. Random seeds (seed=42 for 3-fx SBRT, seed=43 for 5-fx SBRT, seed=44 for 30-fx CFRT) are documented for reproducibility.*

References

1. Keall PJ, Mageras GS, Balter JM, et al. The management of respiratory motion in radiation oncology report of AAPM Task Group 76. Med Phys. 2006;33(10):3874-3900.

2. AAPM Task Group 324 Report 91.b. The Management of Respiratory Motion in Radiation Oncology. Med Phys. 2026. DOI: 10.1002/mp.70677.

3. van Herk M. Errors and margins in radiotherapy. Semin Radiat Oncol. 2004;14(1):52-64.

4. van Herk M, Remeijer P, Rasch C, Lebesque JV. The probability of correct target dosage: dose-population histograms for deriving margins. Int J Radiat Oncol Biol Phys. 2000;47(4):1121-1135.

5. Lujan AE, Larsen EW, Balter JM, Ten Haken RK. A method for incorporating organ motion due to breathing into 3D dose calculations. Med Phys. 1999;26(5):715-720.

6. Seppenwoolde Y, Shirato H, Kitamura K, et al. Precise and real-time measurement of 3D tumor motion in lung due to breathing and heartbeat, measured during radiotherapy. Int J Radiat Oncol Biol Phys. 2002;53(4):822-834.

7. International Commission on Radiation Units and Measurements. Prescribing, Recording, and Reporting Photon-Beam Intensity-Modulated Radiation Therapy (ICRU Report 83). J ICRU. 2010;10(1).

8. Underberg RWM, Lagerwaard FJ, Slotman BJ, Cuijpers JP, Senan S. Use of maximum intensity projections (MIP) for target volume generation in 4DCT scans for lung cancer. Int J Radiat Oncol Biol Phys. 2005;63(1):253-260.

9. Wolthaus JW, Sonke JJ, van Herk M, Belderbos JS, Rossi MM, Lebesque JV, Damen EM. Reconstruction of a time-averaged midposition CT scan for radiotherapy planning of lung cancer patients using deformable registration. Med Phys. 2008;35(9):3998-4011.

10. Sonke JJ, Lebesque J, de Bois J, et al. Frameless stereotactic body radiotherapy for lung cancer using four-dimensional cone beam CT guidance. Int J Radiat Oncol Biol Phys. 2008;70(2):590-598.

11. Shirato H, Shimizu S, Kitamura K, et al. Four-dimensional treatment planning and fluoroscopic real-time tumor-tracking radiotherapy for moving tumor. Int J Radiat Oncol Biol Phys. 2000;48(2):435-442.

12. Vedam SS, Keall PJ, Kini VR, Mostafavi H, Shukla HP, Mohan R. Acquiring a four-dimensional computed tomography dataset using an external respiratory signal. Phys Med Biol. 2003;48(1):45-62.

13. Rietzel E, Chen GT, Choi NC, Willett CG. Four-dimensional image-based treatment planning: Target volume segmentation and dose calculation in the presence of respiratory motion. Int J Radiat Oncol Biol Phys. 2005;61(5):1535-1550.

14. Engelsman M, Sharp GC, Bortfeld T, Onimaru R, Shirato H. How much margin reduction is possible with respiratory gating? Int J Radiat Oncol Biol Phys. 2001;50(5):1357-1365.

15. Guckenberger M, Meyer J, Wilbert J, et al. Precision of image-guided radiotherapy (IGRT) in patient positioning and target volume definition for stereotactic body radiotherapy of lung cancer. Int J Radiat Oncol Biol Phys. 2009;74(2):581-588.

16. Bissonnette JP, Balter PA, Dong L, et al. Quality assurance for image-guided radiation therapy: report of AAPM Task Group 179. Med Phys. 2012;39(4):1946-1963.

17. Wilson EB. Probable inference, the law of large numbers, and test of bias. J Am Stat Assoc. 1927;22(158):209-212.

18. Timmerman R, McGarry M, Yiannoutsos C, et al. Extracranial stereotactic radioablation: results of a phase I study in medically inoperable stage I non-small cell lung cancer. J Clin Oncol. 2006;24(30):4833-4839.

19. Benedict SH, Yenice KM, Followill D, et al. Stereotactic body radiation therapy: the report of AAPM Task Group 101. Med Phys. 2010;37(8):4078-4101.

20. Stroom JC, de Boer HC, Huizenga H, Visser AG. Inclusion of geometrical uncertainties in radiotherapy treatment planning by means of coverage probability. Int J Radiat Oncol Biol Phys. 1999;43(4):905-919.